

**CONTEXT:** This study had the aim to reconstruct airflows as it has been needed to solve an outnumber of agricultural engineering problems. Airflows are crucial from drying, wetting and refreshing air to removal of particles and gases. However, physical laws governing airflows should address turbulent conditions, requiring intensive use of computational resources in order to get a reliable visualization.

**OBJECTIVES:** 1) Train a physics informed diffusion model (PIDM) with data from a computational fluid dynamics (CFD) model to reconstruct airflows subjected to turbulent conditions.  
2) Use minimum input data in the PIDM to reconstruct airflows.  
3) Assess the PIDM performance.

## 1 PROBLEM DESCRIPTION

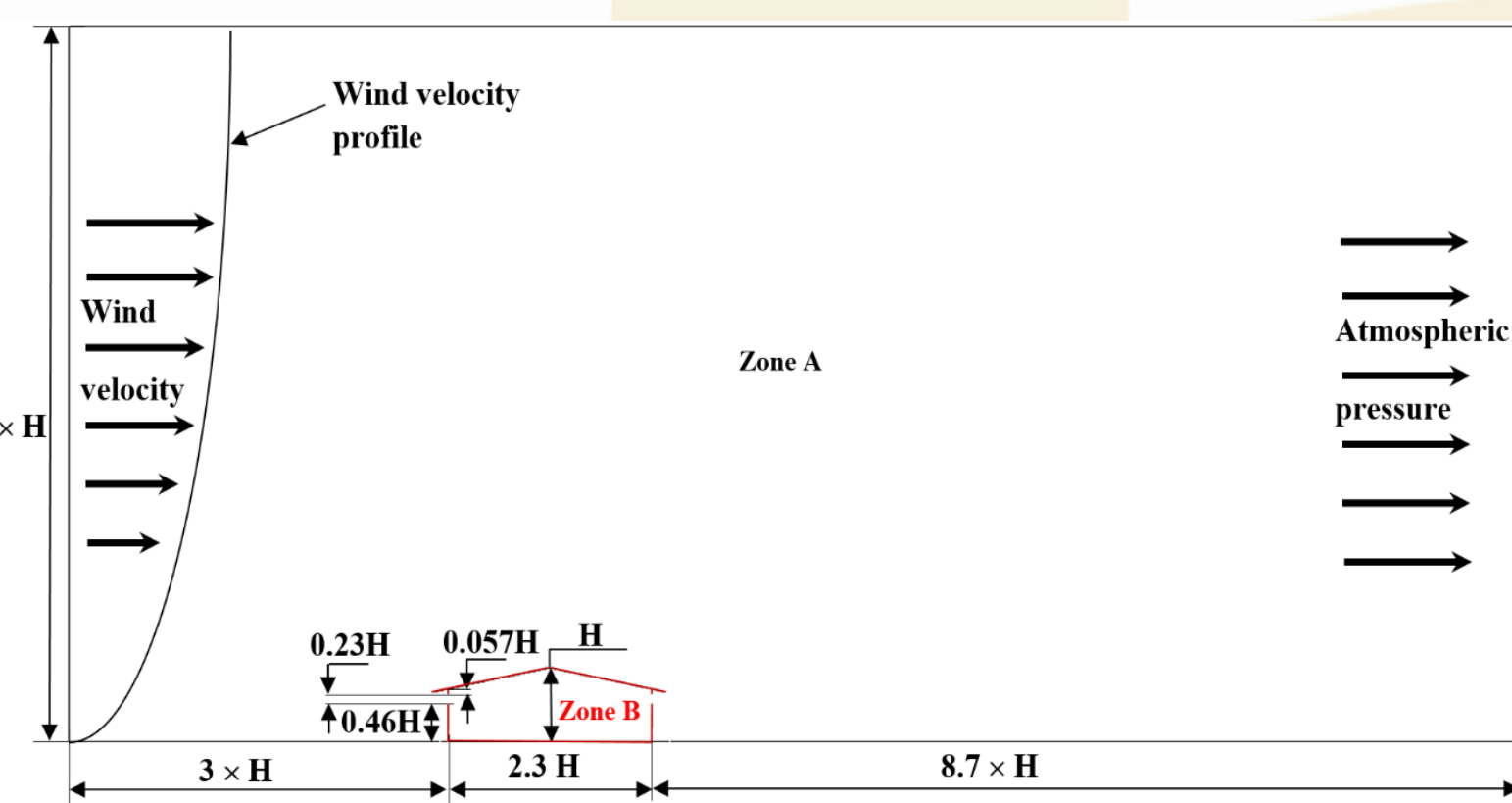
- A two-dimensional representation of an agricultural building relying on natural ventilation was the baseline for a CFD model.
- Cross-ventilation was simulated (1-12 m/s) in the CFD model with the aim to assess internal air renewal with fresh external air.
- CFD model was discretized to follow the approach of finite volume where Navier-Stokes equations were applied.
- Governing equations of conservation of mass and momentum equation deduce the velocity field  $u$ , being  $u_1$  and  $u_2$  the horizontal ( $x$ ) and vertical ( $y$ ) velocities components:

$$\frac{\partial u_1}{\partial x} + \frac{\partial u_2}{\partial y} = 0$$

$$\rho(u \cdot \nabla)u = -\nabla p + \nabla \cdot (\mu_{eff} \nabla u) + f$$

where  $\rho$  and  $\mu_{eff}$  are air density and effective viscosity.  $\nabla \cdot$  is a divergence operator, for horizontal velocity as  $\frac{\partial}{\partial x}(\mu_{eff} \frac{\partial u_1}{\partial x})$  and  $u \cdot \nabla$  is a gradient operator, for horizontal velocity as  $u_1 \frac{\partial u_1}{\partial x} + u_2 \frac{\partial u_1}{\partial y}$ . Terms  $p$  and  $f$  are pressure and external forces.

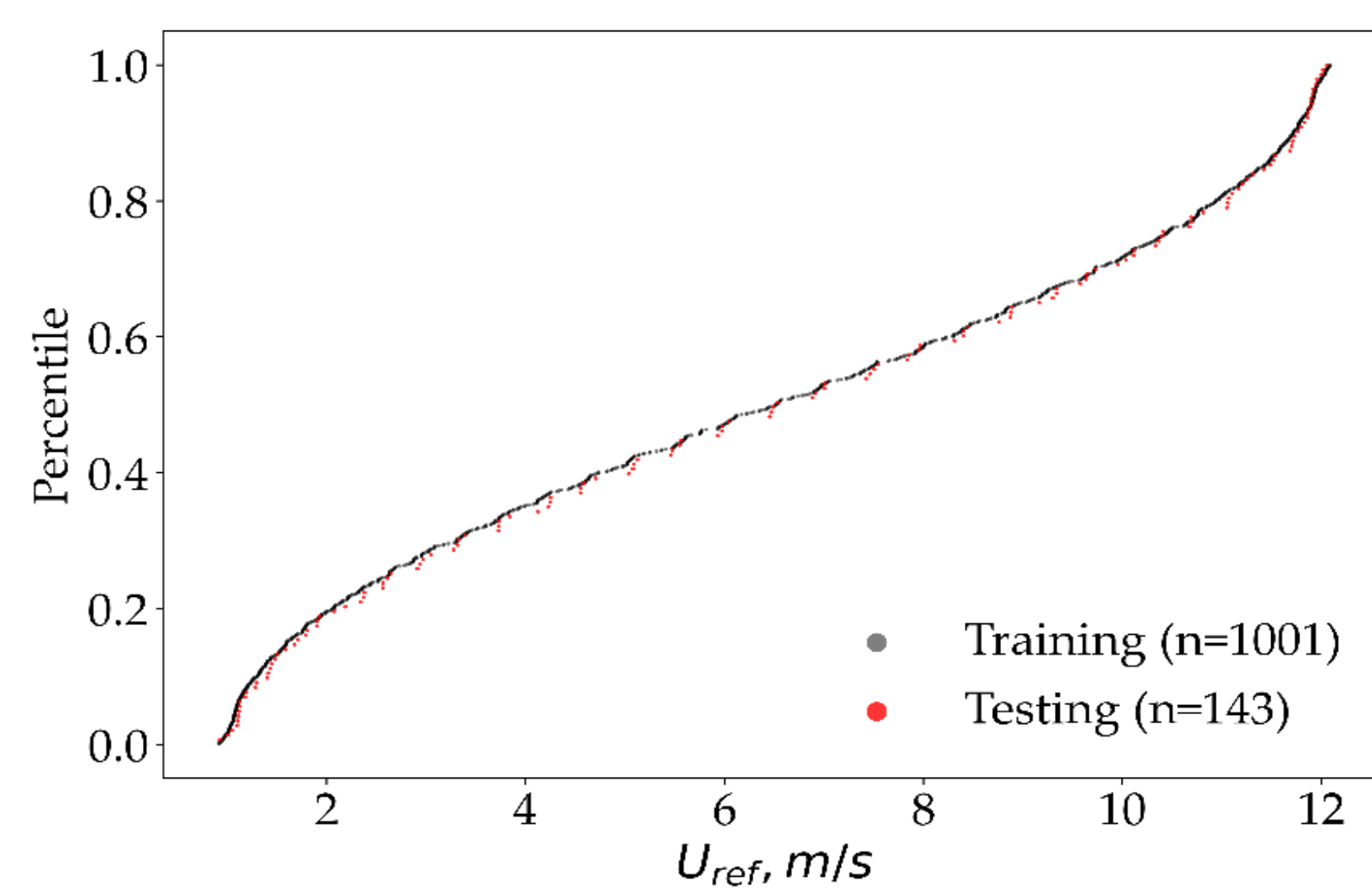
- Numerical solutions of air velocity from the 2D CFD model addressed specific features representing the building within a domain that followed a Standard  $\kappa$ - $\epsilon$  turbulence model.



Domain and boundaries specifications of the vicinity (Zone A) of a simplified agricultural building (Zone B).

## 2 CFD DATA

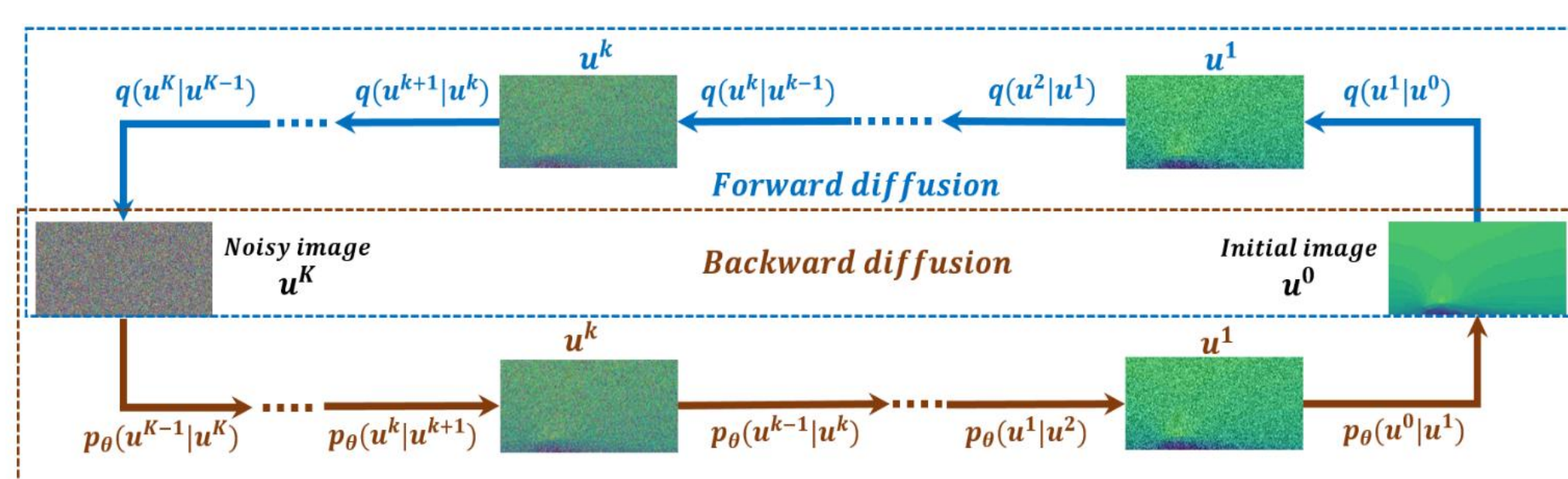
- Air velocity from 1144 CFD numerical solutions were retrieved.
- Our PIDM was trained and tested with 87% and 13% of data, respectively.



Uref values used at the velocity inlet of the 2D CFD model to find the numerical solutions defining the training and testing datasets.

## 3 DEVELOPMENT OF A PHYSICS INFORMED DIFFUSION MODEL

- It relies on the Denoising Diffusion Probabilistic Model (DDPM) which consists of systematically adding noise to an image.
- A DDPM includes two stages: Forward diffusion and backward (or reverse) diffusion.
- Forward diffusion will add Gaussian noise to the image, typically in 1000 steps.
- Backward diffusion will remove noise added to the image.



Training configuration of the DDPM through forward and backward diffusion process.

## PHYSICS INFORMED CONDITION

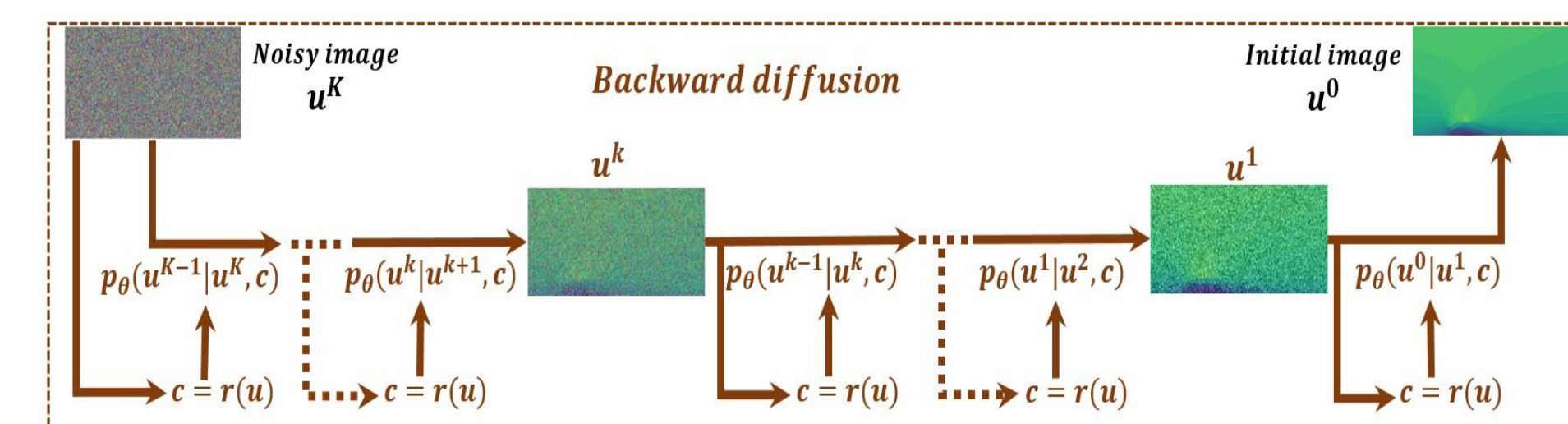
- Additional information is provided to the DDPM through reanalysis of the velocity  $u$  to find a residual  $c=r(u)$  deduced from the momentum equation:

$$r(u) = \rho(u \cdot \nabla)u + (1 - \alpha)\nabla p - \nabla \cdot (\mu_{eff} \nabla u)$$

term  $\nabla p$  is estimated from the Poisson equation whereas  $f$  is found in function of  $\nabla p$  through  $\alpha$ . Poisson equation is:

$$\nabla^2 p = -\nabla[\rho(u \cdot \nabla)u - \nabla \cdot (\mu_{eff} \nabla u)]$$

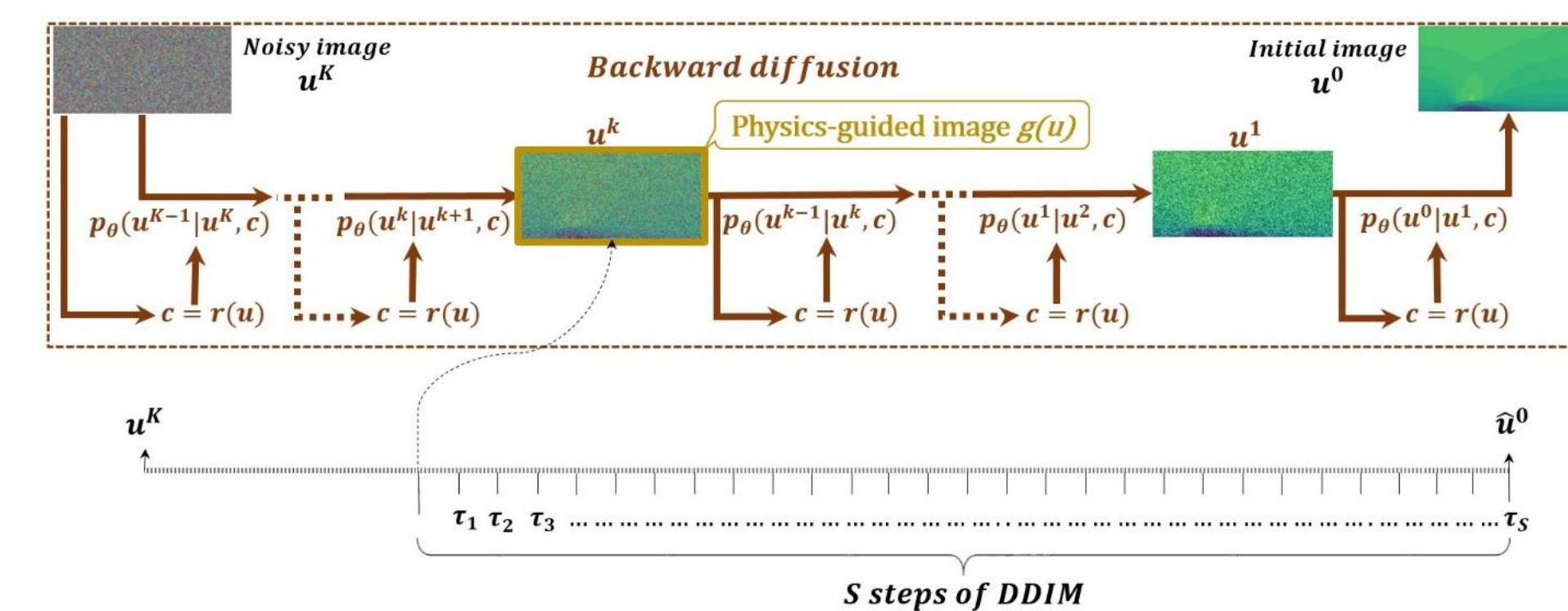
- Effective viscosity  $\mu_{eff} = \mu + \rho C_\mu \kappa^2 / \epsilon$  where  $\mu$  is dynamic viscosity,  $C_\mu$  is a constant of the  $\kappa$ - $\epsilon$  model.  $\kappa$  is the turbulence kinetic energy and  $\epsilon$  is the rate of dissipation. Both  $\kappa$  and  $\epsilon$  were retrieved from the CFD model.



Incorporating residuals to the steps of the DDPM.

## SPEEDING UP IMAGE RECONSTRUCTION WITH DENOISING DIFFUSION IMPLICIT MODEL (DDIM)

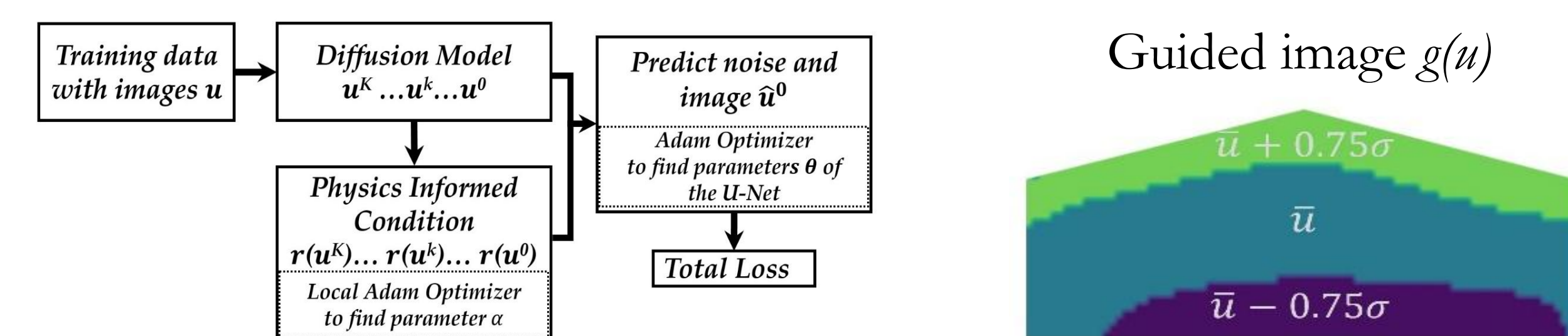
- Use of a guided image
- Skipping steps with DDIM for generating an image.



Backward diffusion with a guided image  $g(u)$  to reconstruct an image in  $S$  steps.

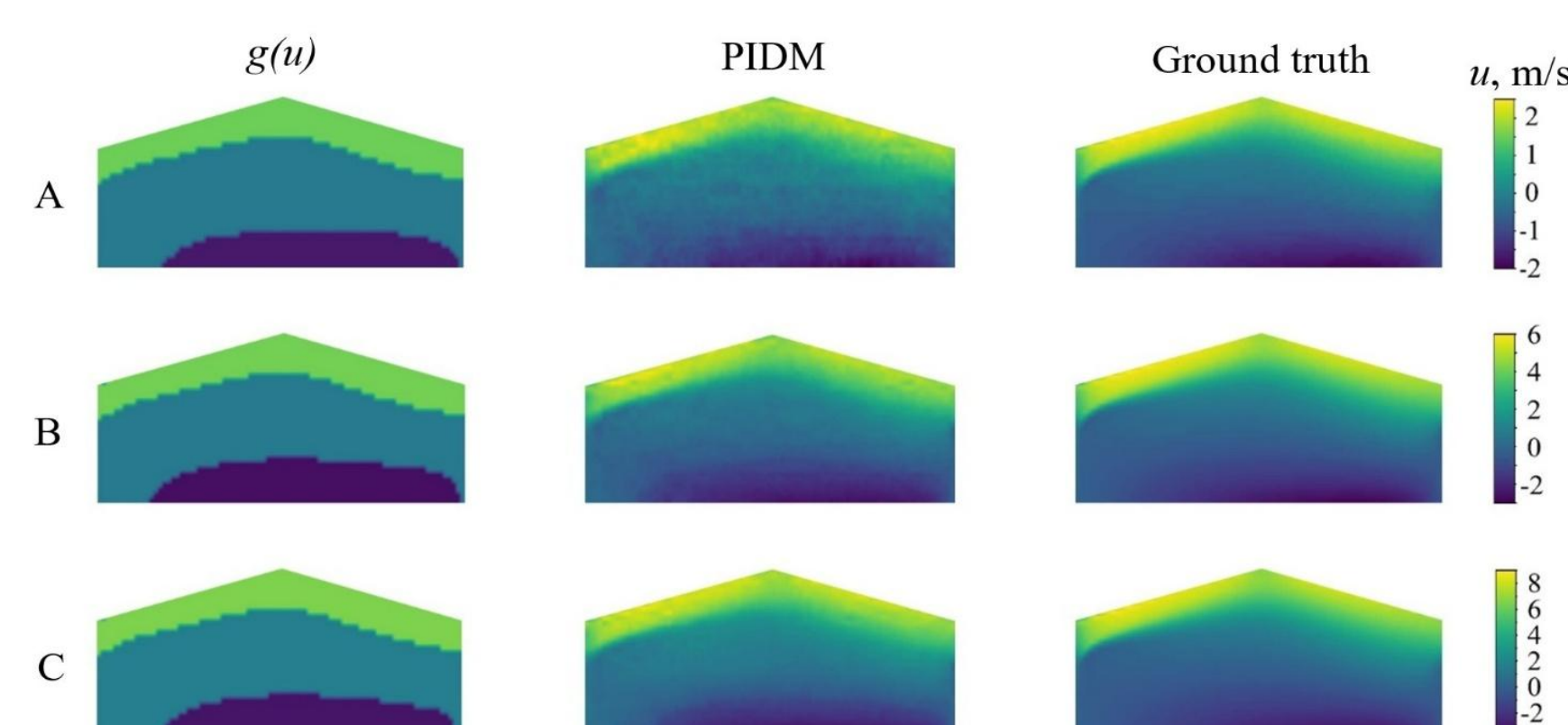
## PUTTING A PIDM INTO PRACTICE

- Training relies on calibration of the U-Net neural network.
- Testing needs a guided image  $g(u)$  as input data for the PIDM, where  $\bar{u}$  and  $\sigma$  were the mean and standard deviation of the velocity.

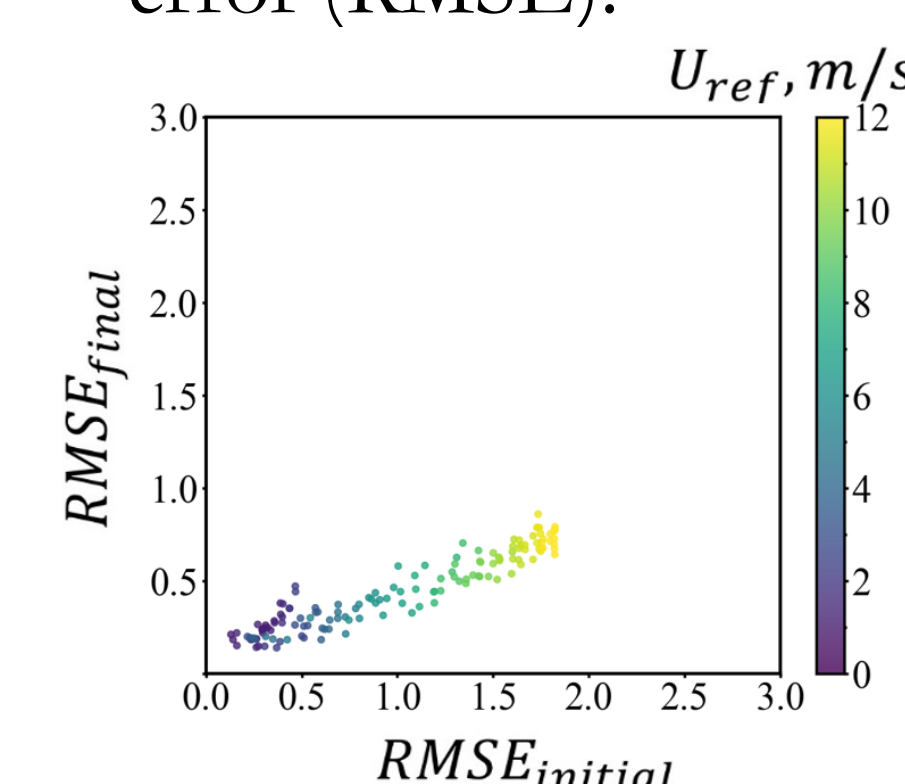


## RESULTS

- Three cases of physics guided image  $g(u)$  reconstructed by the PIDM



- Assessing 143 reconstructed images by the PIDM through the root mean square error (RMSE).



Initial and final values of the RMSE correspond to the guided image  $g(u)$  and reconstructed image by the PIDM.

## FURTHER INFORMATION

